Early-Pandemic COVID-19 Epidemiologic Analysis

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1 Study Data

This study uses a 10,000-record sample of reported COVID-19 cases restricted to the early pre-vaccine study period from March 15 through December 10, 2020.

```r
# Load COVID-19 surveillance sample
covid <- read.csv(
    "Data/covid_early_pandemic_sample_10k.csv",
    stringsAsFactors = FALSE
)

dim(covid)
```

```
[1] 10000    12
```

```r
names(covid)
```

```
 [1] "cdc_case_earliest_dt"    "cdc_report_dt"
 [3] "onset_dt"                "current_status"
 [5] "sex"                     "age_group"
 [7] "race_ethnicity_combined" "hosp_yn"
 [9] "icu_yn"                  "death_yn"
[11] "medcond_yn"              "pos_spec_dt"
```

```r
head(covid)
```

```
     cdc_case_earliest_dt           cdc_report_dt                 onset_dt
1 2020-03-27T00:00:00.000 2021-12-06T00:00:00.000 2020-03-27T00:00:00.000
2 2020-03-27T00:00:00.000 2020-04-22T00:00:00.000 2020-03-27T00:00:00.000
3 2020-03-27T00:00:00.000 2020-04-05T00:00:00.000 2020-03-27T00:00:00.000
4 2020-03-27T00:00:00.000 2020-03-27T00:00:00.000 2020-03-27T00:00:00.000
5 2020-03-27T00:00:00.000 2020-06-05T00:00:00.000 2020-03-27T00:00:00.000
6 2020-03-27T00:00:00.000
             current_status  sex     age_group race_ethnicity_combined hosp_yn
1 Laboratory-confirmed case Male 40 - 49 Years     White, Non-Hispanic     Yes
2 Laboratory-confirmed case Male 40 - 49 Years     White, Non-Hispanic      No
3 Laboratory-confirmed case Male 40 - 49 Years     White, Non-Hispanic      No
4 Laboratory-confirmed case Male 40 - 49 Years     White, Non-Hispanic     Yes
5 Laboratory-confirmed case Male 40 - 49 Years     White, Non-Hispanic      No
6 Laboratory-confirmed case Male 40 - 49 Years     White, Non-Hispanic      No
   icu_yn death_yn medcond_yn             pos_spec_dt
1      No       No        Yes 2020-03-29T00:00:00.000
2 Missing       No        Yes 2020-03-27T00:00:00.000
3 Missing       No    Missing 2020-03-28T00:00:00.000
4 Missing  Missing    Missing
5 Missing  Missing        Yes 2020-04-08T00:00:00.000
6 Missing  Missing    Missing
```

2 Analytic Cohort

The analysis is restricted to cases with known exposure and outcome status. Exposure is defined by `medcond_yn`, indicating whether a pre-existing medical condition was reported. The outcome is defined by `hosp_yn`, indicating a hospitalization was reported.

`medcond_yn` and `hosp_yn` are coded as {`Yes`, `No`, `Unknown`, `Missing`}. Only `Yes` and `No` records for either the exposure or outcome are kept for the complete-case analysis.

```
analysis <- covid[
    covid$medcond_yn %in% c("Yes", "No") &
    covid$hosp_yn %in% c("Yes", "No"),
]

dim(analysis)
```

```
[1] 1464   12
```

The resulting complete-case analytic cohort contains 1,464 cases.

3 Crude Association

3.1 Exposure-Outcome Table

We first estimate the crude association between the exposure (reported pre-existing medical condition) and outcome (hospitalization).

```
exposure_outcome <- table(
    analysis$medcond_yn,
    analysis$hosp_yn
)

exposure_outcome
```

```
      No Yes
  No  728  27
  Yes 459 250
```

The table's orientation lists exposure as rows and outcome as columns.

3.2 Risk Ratio and Risk Difference

Hospitalization risk is estimated separately among exposed and unexposed cases.

```
# Extract cells from exposure-outcome table
a <- exposure_outcome["Yes", "Yes"]
b <- exposure_outcome["Yes", "No"]
c <- exposure_outcome["No", "Yes"]
d <- exposure_outcome["No", "No"]

# Hospitalization risks
risk_exposed <- a / (a + b)
risk_unexposed <- c / (c + d)

# Effect measures
risk_ratio <- risk_exposed / risk_unexposed
risk_difference <- risk_exposed - risk_unexposed

risk_exposed
```

```
[1] 0.3526093
```

```
risk_unexposed
```

```
[1] 0.03576159
```

```
risk_ratio
```

```
[1] 9.860001
```

```
risk_difference
```

```
[1] 0.3168477
```

Among cases in the complete-case analytic cohort, hospitalization risk was approximately 35.3% among those with a reported pre-existing medical condition and 3.6% among those without one. The crude risk ratio was 9.86, corresponding to an absolute risk difference of approximately 31.7 percentage points.

3.3 95% Confidence Interval for Risk Ratio

The conventional Wald confidence interval for the risk ratio is defined as:

$$SE[\ln(RR)] = \sqrt{\frac{1}{a} - \frac{1}{a+b} + \frac{1}{c} - \frac{1}{c+d}}$$

```
se_log_rr <- sqrt(
    (1 / a) - (1 / (a + b)) +
    (1 / c) - (1 / (c + d))
)

log_rr <- log(risk_ratio)

log_rr_lower <- log_rr - (1.96 * se_log_rr)
log_rr_upper <- log_rr + (1.96 * se_log_rr)

rr_lower <- exp(log_rr_lower)
rr_upper <- exp(log_rr_upper)

c(
    RR = risk_ratio,
    Lower_95_CI = rr_lower,
    Upper_95_CI = rr_upper
)
```

```
       RR Lower_95_CI Upper_95_CI
 9.860001    6.718710   14.469983
```

The crude risk ratio was 9.86 (95% CI: 6.72–14.47).

3.4 95% Confidence Interval for Risk Difference

For two independent proportions the standard error is defined as:

$$SE(RD) = \sqrt{\frac{R_1(1-R_1)}{n_1} + \frac{R_0(1-R_0)}{n_0}}$$

```r
n_exposed <- a + b
n_unexposed <- c + d

se_rd <- sqrt(
    (risk_exposed * (1 - risk_exposed) / n_exposed) +
    (risk_unexposed * (1 - risk_unexposed) / n_unexposed)
)

rd_lower <- risk_difference - (1.96 * se_rd)
rd_upper <- risk_difference + (1.96 * se_rd)

c(
    RD = risk_difference,
    Lower_95_CI = rd_lower,
    Upper_95_CI = rd_upper
)
```

```
        RD Lower_95_CI Upper_95_CI
 0.3168477   0.2792667   0.3544287
```

The crude risk difference was 31.7 percentage points (95% CI: 27.9–35.4 percentage points).

These estimates are crude and unadjusted. These confidence intervals treat observations as independent.

4 Descriptive Analysis

Age and sex were hypothesized to be potential confounders based on the DAG and literature review. Consequently, their distributions are described according to both exposure and outcome status before adjusted analysis.

4.1 Age

4.1.1 Age by Exposure

```r
age_exposure <- table(
    analysis$age_group,
    analysis$medcond_yn
)
```

```r
age_exposure
```

```
                No Yes
  0 - 9 Years   134  15
  10 - 19 Years  65  23
  20 - 29 Years 171  78
  30 - 39 Years 164  98
  40 - 49 Years  52  46
  50 - 59 Years 104  92
  60 - 69 Years  51 224
  70 - 79 Years   9  49
  80+ Years       5  84
```

```r
round(
    prop.table(age_exposure, margin = 2) * 100,
    1
)
```

```
                 No  Yes
  0 - 9 Years   17.7  2.1
  10 - 19 Years  8.6  3.2
  20 - 29 Years 22.6 11.0
  30 - 39 Years 21.7 13.8
  40 - 49 Years  6.9  6.5
  50 - 59 Years 13.8 13.0
  60 - 69 Years  6.8 31.6
  70 - 79 Years  1.2  6.9
  80+ Years      0.7 11.8
```

The column percentages describe the age distribution within each exposure group, $P(Age \mid E)$.

4.1.2 Hospitalization by Age

```r
age_outcome <- table(
    analysis$age_group,
    analysis$hosp_yn
)

age_outcome
```

```
                No Yes
  0 - 9 Years   145   4
  10 - 19 Years  86   2
  20 - 29 Years 240   9
  30 - 39 Years 238  24
  40 - 49 Years  87  11
  50 - 59 Years 184  12
  60 - 69 Years 127 148
  70 - 79 Years  27  31
  80+ Years      53  36
```

```r
round(
    prop.table(age_outcome, margin = 1) * 100,
    1
)
```

```
                 No  Yes
  0 - 9 Years   97.3  2.7
  10 - 19 Years 97.7  2.3
  20 - 29 Years 96.4  3.6
  30 - 39 Years 90.8  9.2
  40 - 49 Years 88.8 11.2
  50 - 59 Years 93.9  6.1
  60 - 69 Years 46.2 53.8
  70 - 79 Years 46.6 53.4
  80+ Years     59.6 40.4
```

The row percentages describe hospitalization status within each age group, $P(Y \mid Age)$.

4.2 Sex

4.2.1 Sex by Exposure

```
sex_exposure <- table(
    analysis$sex,
    analysis$medcond_yn
)

sex_exposure
```

```
           No Yes
  Female  368 327
  Male    386 381
  Missing   1   0
  Unknown   0   1
```

```
round(
    prop.table(sex_exposure, margin = 2) * 100,
    1
)
```

```
            No  Yes
  Female  48.7 46.1
  Male    51.1 53.7
  Missing  0.1  0.0
  Unknown  0.0  0.1
```

The column percentages describe the sex distribution within each exposure group, $P(Sex \mid E)$.

4.2.2 Hospitalization by Sex

```r
sex_outcome <- table(
    analysis$sex,
    analysis$hosp_yn
)

sex_outcome
```

```
           No Yes
  Female  573 122
  Male    612 155
  Missing   1   0
  Unknown   1   0
```

```r
round(
    prop.table(sex_outcome, margin = 1) * 100,
    1
)
```

```
             No   Yes
  Female   82.4  17.6
  Male     79.8  20.2
  Missing 100.0   0.0
  Unknown 100.0   0.0
```

The row percentages describe hospitalization status within each sex group, $P(Y \mid Sex)$.

5 Adjusted Analysis

The adjusted analysis needs to account for the age and sex variables because they were identified as potential confounders. Two records in the complete-case analytic cohort have sex coded as `Missing` or `Unknown` and have been excluded from the analysis.

```r
adjusted_analysis <- analysis[
    analysis$sex %in% c("Female", "Male"),
]

dim(adjusted_analysis)
```

```
[1] 1462   12
```

```
table(adjusted_analysis$sex)
```

```
Female   Male
   695    767
```

```
table(adjusted_analysis$age_group)
```

```
  0 - 9 Years 10 - 19 Years 20 - 29 Years 30 - 39 Years 40 - 49 Years
          149            88           248           262            98
50 - 59 Years 60 - 69 Years 70 - 79 Years     80+ Years
          196           275            57            89
```

```
# Encode variables for regression
adjusted_analysis$hospitalized <- factor(
    adjusted_analysis$hosp_yn,
    levels = c("No", "Yes")
)

adjusted_analysis$medcond <- factor(
    adjusted_analysis$medcond_yn,
    levels = c("No", "Yes")
)

adjusted_analysis$sex_factor <- factor(
    adjusted_analysis$sex,
    levels = c("Female", "Male")
)

adjusted_analysis$age_factor <- factor(
    adjusted_analysis$age_group,
    levels = c(
        "0 - 9 Years",
        "10 - 19 Years",
        "20 - 29 Years",
        "30 - 39 Years",
        "40 - 49 Years",
        "50 - 59 Years",
```

```
        "60 - 69 Years",
        "70 - 79 Years",
        "80+ Years"
    )
)

levels(adjusted_analysis$hospitalized)
```

```
[1] "No"  "Yes"
```

```
levels(adjusted_analysis$medcond)
```

```
[1] "No"  "Yes"
```

```
levels(adjusted_analysis$sex_factor)
```

```
[1] "Female" "Male"
```

```
levels(adjusted_analysis$age_factor)
```

```
[1] "0 - 9 Years"   "10 - 19 Years" "20 - 29 Years" "30 - 39 Years"
[5] "40 - 49 Years" "50 - 59 Years" "60 - 69 Years" "70 - 79 Years"
[9] "80+ Years"
```

```
table(adjusted_analysis$hospitalized)
```

```
  No  Yes
1185  277
```

```
table(adjusted_analysis$medcond)
```

```
 No Yes
754 708
```

```r
table(adjusted_analysis$sex_factor)
```

```
Female   Male
   695    767
```

```r
table(adjusted_analysis$age_factor)
```

```
  0 - 9 Years 10 - 19 Years 20 - 29 Years 30 - 39 Years 40 - 49 Years
          149            88           248           262            98
50 - 59 Years 60 - 69 Years 70 - 79 Years     80+ Years
          196           275            57            89
```

The resulting adjusted-analysis cohort contained 1,462 cases. Hospitalization was modeled with No as the reference category and Yes as the event. Reported pre-existing medical condition used No as the reference category, sex used Female as the reference category, and age group was modeled categorically with 0–9 Years as the reference category.

5.1 Logistic Regression

A multivariable logistic regression model was fit with hospitalization as the outcome, reported pre-existing medical condition as the exposure, and age group and sex as prespecified adjustment variables.

The logistic model estimates conditional log odds of hospitalization. Exponentiated coefficients therefore represent adjusted odds ratios rather than adjusted risk ratios.

```r
logistic_model <- glm(
    hospitalized ~ medcond + age_factor + sex_factor,
    data = adjusted_analysis,
    family = binomial(link = "logit")
)

summary(logistic_model)
```

```
Call:
glm(formula = hospitalized ~ medcond + age_factor + sex_factor,
```

```
    family = binomial(link = "logit"), data = adjusted_analysis)

Coefficients:
                          Estimate Std. Error z value Pr(>|z|)
(Intercept)             -4.180e+00  5.354e-01  -7.807 5.86e-15 ***
medcondYes               1.874e+00  2.319e-01   8.079 6.50e-16 ***
age_factor10 - 19 Years -5.267e-01  8.929e-01  -0.590 0.555311
age_factor20 - 29 Years -1.615e-01  6.273e-01  -0.258 0.796792
age_factor30 - 39 Years  6.146e-01  5.667e-01   1.084 0.278178
age_factor40 - 49 Years  7.327e-01  6.196e-01   1.183 0.237001
age_factor50 - 59 Years  2.942e-05  6.066e-01   0.000 0.999961
age_factor60 - 69 Years  2.624e+00  5.424e-01   4.838 1.31e-06 ***
age_factor70 - 79 Years  2.567e+00  5.977e-01   4.295 1.75e-05 ***
age_factor80+ Years      1.964e+00  5.850e-01   3.358 0.000785 ***
sex_factorMale           2.805e-01  1.824e-01   1.538 0.123996
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 1419.45  on 1461  degrees of freedom
Residual deviance:  947.29  on 1451  degrees of freedom
AIC: 969.29

Number of Fisher Scoring iterations: 6
```

5.1.1 Logistic regression limitations

Our outcome is binary, so logistic regression is an obvious modeling choice. However, the logit link models the log odds of the outcome rather than the risk directoy. The fitted model can be written as:

$$\text{logit}(p_i) = \beta_0 + \beta_E E_i + \sum_{k=1}^{8} \beta_{A_k} A_{ik} + \beta_S S_i,$$

where

$$p_i = P(Y_i = 1 \mid E_i, A_i, S_i).$$

Here, E_i represents reported pre-existing medical condition stuatus, A_{ik} represents the eight indicator variables from the non-reference age groups, and S_i represents sex.

The exposure coefficient β_E is estimated on the log-odds scale. Exponentiating this coefficient gives the adjusted odds ratio:

$$OR_{\text{adjusted}} = e^{\beta_E},$$

rather than an adjusted risk ratio, which was selected as teh primary measure for this analysis. Consequently, the adjusted risk ratio will be estimated from the fitted logistic regression model using regression standardization.